

- Q1** The theory of Lamarckism states that:
- (A) new species arise due to genetic mutations.
 - (B) organisms evolve through the inheritance of acquired characteristics.
 - (C) evolution is driven by natural selection.
 - (D) organisms evolve through genetic drift.

- Q2** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Flippers of Penguins and Dolphins are homologous organs.

Reason R: Flippers of Penguins and Dolphins are example of convergent evolution.

In the light of the above statements, choose the **correct** answer from the options given below:

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

- Q3** The concept of homologous structures can be used to infer that:

- (A) different organisms with similar functions must have evolved from a common ancestor.
- (B) organisms sharing homologous structures are unrelated but adapted to similar environments.
- (C) anatomical similarity of organisms suggests that they might share a common ancestry, though they have diverged over time.
- (D) organisms sharing homologous structures evolved separately with no common ancestors.

- Q4** Given below are two statements: one is labelled as Assertion A and the other is

labelled as Reason R:

Assertion A: Karl Ernst von Baer disapproved of Haeckel's proposal regarding embryological support for evolution.

Reason R: He noted that embryos do not pass through the adult stages of other animals.

In the light of the above statements, choose the **correct** answer from the options given below:

- (A) A is true but R is false.
- (B) A is false but R is true.
- (C) Both A and R are true and R is the correct explanation of A.
- (D) Both A and R are true and R is NOT the correct explanation of A.

- Q5** Choose the wrong pair

- (A) Divergent evolution - Fore limbs of whales, bat, cheetah and human
- (B) Convergent evolution - Flippers of penguins and dolphins
- (C) Homologous structures - Vertebrate hearts
- (D) Analogous structures - Tendril of Bougainvillea and Cucurbita

- Q6** Read the following statements and choose incorrect one.

- (A) Fossils are remains of hard parts of life forms found in rocks
- (B) First form of life arose slowly through evolutionary forces from non-living molecules, is known as biogenesis.
- (C) Wings of butterfly and birds shows homology.
- (D) Evolution is not a directed process in the sense of determination.

- Q7** Which of the following is not a reptilian character of Archaeopteryx ?

- (A) The bones are not pneumatic



- (B) Presence of weak sternum
- (C) An intimate fusion of skull bones
- (D) Jaws bear teeth

Q8 Select the **odd** one out w.r.t the forelimbs of whale, bat, cheetah and humans.

- A. Similarities in the pattern of bones.
- B. Analogous organs
- C. Convergent evolution
- D. Divergent evolution
- E. Different functions performed by similar structures.

Choose the most appropriate answer from the options given below:

- (A) A, B and C only
- (B) A, D and E only
- (C) A and D only
- (D) A, D and C only

Q9 Read the following statements (**A – E**)

- (A) Presence of a short tail in newborn babies is an example of atavism

- (B) The rate of appearance of new forms has no link to the life span
- (C) Those animals which are better fit undergo natural selection
- (D) Evolution is not a directed process in the sense of determinism
- (E) According to Darwin, evolution is a rapid, singlestep process

How many of above statements are **correct**?

- (A) Two
- (B) Three
- (C) Four
- (D) Five

Q10 Gill clefts and notochord appear in the embryonic development of all vertebrates from fishes to mammals. This supports the theory of:

- (A) recapitulation.
- (B) metamorphosis.
- (C) biogenesis.
- (D) abiogenesis.



Answer Key

Q1 B
Q2 D
Q3 C
Q4 C
Q5 D

Q6 C
Q7 C
Q8 B
Q9 B
Q10 A



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

(2)

The theory of Lamarckism states that organisms evolve through the inheritance of acquired characteristics.

(NEW NCERT 12th Page No. 118)

Video Solution:



Q2 Text Solution:

(4)

Flippers of Penguins and Dolphins have different structure but they perform same function. They are analogous structure and are a result of convergent evolution.

Video Solution:



Q3 Text Solution:

(3)

- Homology indicates common ancestry.
- Homologous structures developed in different directions due to adaptations to different needs in different organisms (divergent evolution).

[New NCERT Class 12th Page No. 114]

Video Solution:



Q4 Text Solution:

- **Assertion A is true** because **Karl Ernst von Baer disapproved of Haeckel's proposal regarding embryological support for evolution**. Haeckel's **biogenetic law** stated that "ontogeny recapitulates phylogeny," meaning that embryos pass through stages resembling adult forms of their evolutionary ancestors. Von Baer rejected this idea.
- **Reason R is true** because **von Baer noted that embryos do not pass through the adult stages of other animals**. Instead, he proposed that embryos develop progressively, starting from a general form shared among related species and then differentiating into species-specific traits.
- **R correctly explains A** because von Baer's observations provided an alternative explanation, leading to the rejection of Haeckel's concept.

Video Solution:



Q5 Text Solution:

Tendrils of cucurbita and thorns of bougainvillea are example of homology. These structures are evolved from axillary bud.



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Tendrils function in climbing.

Thorns function in protection and reducing transpiration.

Homologous organs are those that have similar structure but have different functions.

Video Solution:



Q6 Video Solution:



Q7 Video Solution:



Q8 Text Solution:
(2)

- Whales, bats, Cheetah and human (all mammals) share similarities in the pattern of bones of forelimbs.
- Though these forelimbs perform different functions in these animals, they have similar anatomical structure – all of them have humerus, radius, ulna, carpals, metacarpals and phalanges in their forelimbs.
- Hence, in these animals, the same structure developed along different directions due to adaptations to different needs.

- This is divergent evolution and these structures are homologous.
- Homology indicates common ancestry. Other examples are vertebrate hearts or brains.

[New NCERT Class 12th Page No. 114]

Video Solution:



Q9 Video Solution:



Q10 Text Solution:
(1)

- According to the recapitulation theory that development from egg to adult repeats or recapitulates in a condensed form the stages through which ancestors have passed in the course of their evolution.
- In other words, it can be said that ontogeny repeats phylogeny.
- Frog has in its development an aquatic, fish-like tadpole larva.
- The larva has a tail with a fin for swimming and gills for breathing in water.
- These stages indicate that the frog has evolved from a fish-like ancestor.

Video Solution:



[Android App](#)

| [iOS App](#)

| [PW Website](#)